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Hybrid Particle Swarm Optimization With Genetic Algorithm to Train Artificial Neural Networks for Short-Term Load Forecasting

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ABSTRACT

This research proposes a new training algorithm for artificial neural networks (ANNs) to improve the short-term load forecasting (STLF) performance. The proposed algorithm overcomes the so-called training issue in ANNs, where it traps in local minima, by applying genetic algorithm operations in particle swarm optimization when it converges to local minima. The training ability of the hybridized training algorithm is evaluated using load data gathered by Electricity Generating Authority of Thailand. The ANN is trained using the new training algorithm with one-year data to forecast equal 48 periods of each day in 2013. During the testing phase, a mean absolute percentage error (MAPE) is used to evaluate performance of the hybridized training algorithm and compare them with MAPEs from Backpropagation, GA, and PSO. Yearly average MAPE and the average MAPEs for weekdays, Mondays, weekends, Holidays, and Bridging holidays show that PSO+GA algorithm outperforms other training algorithms for STLF.

KEYWORDS

Artificial Neural Network, Back-Propagation, Genetic Algorithm, Particle Swarm Optimization, Short-Term Load Forecasting

INTRODUCTION

STLF is vital for many utility activities: security analysis, fuel purchasing, generator scheduling, and maintenance scheduling (Rana & Koprinska, 2016). Forecasting errors can negatively affect the monetary savings regardless of whether its positive or negative. Over forecasting causes producing or purchasing extra electricity. Under forecasting causes revenue reduction by not having of enough electricity to cover the customer demand (Ismail, Yahya, & Mahpol, 2009). However, STLF is difficult due to the nonlinear behavior of the load consumption. Load consumption depends on many factors such as weather conditions, human behavior, and commercial and social activities (Singh & Singh, 2001). Therefore, contemplating the significance and the difficulty of STLF, a number of Artificial Intelligence (AI) and statistical based techniques have been suggested by researchers (Hassan, Khosravi, & Jaafar, 2015). Even though, statistical based forecasting techniques such as Regression

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Analysis (Dudek, 2016) and Stochastic Time Series (Clements, Hurn, & Li, 2016) abound in STLF, these are good at forecasting linear time series. Since electricity load consumption is highly nonlinear, AI based forecasting techniques such as Deep Learning (Coelho et al., 2016; Dedinec, Filiposka, Dedinec, & Kocarev, 2016; Qiu, Ren, Suganthan, & Amaratunga, 2017), Fuzzy Logic (Hassan, Khosravi, Jaafar, & Khanesar, 2016), machine learning based models (Jurado, Nebot, Mugica, & Avellana, 2015), and Artificial Neural Networks (ANNs) (Chae, Horesh, Hwang, & Lee, 2016; Ding, Benoit, Foggia, Bésanger, & Wurtz, 2016) are better for STLF.

ANN is introduced to STLF as a pattern matching technique. Recognizing and learning nonlinear patterns in data and adjusting its weights based on them to work with unseen data is the principal operation of ANN. Due to its continues accuracy for STLF, use of ANN is ample in the field. However, forecasting capabilities of ANN have been proved by comparing it with other forecasting techniques: Gajowniczek and Ząbkowski (Gajowniczek & Ząbkowski, 2014) use Multi-Layer Perceptron (MLP) and Support Vector Machines (SVM) in their research for electricity forecasting and show that MLP performs better compared to SVM. ANN and Box-Jenkins methods for electricity forecasting are compared by (Ramakrishna, Boiroju, & Reddy, 2011). Results of their research show that ANN performs better in electricity demand forecasting than Box-Jenkins method. Ringwood et et., (Ringwood, Bofelli, & Murray, 2001) also use ANN for forecasting Irish electricity demand. They find that ANN is better for forecasting all three time horizons: Short-Term Load Forecasting (STLF) for next hour to one week, Medium-Term Load Forecasting (MTLF) for one week to one month, and Long-Term Load Forecasting (LTLF) for one months to several years.

However, ANN has some limitations and continues monitoring is required to minimize their effect on forecasting outcomes (Singh & Singh, 2001): ANN takes large computational times as it requires many training cycles to learn the patterns in large datasets, forecasts depend on the random initial values, difficulty of selecting inputs and targets, and determining the number of hidden units (layers and neurons). One of the most highlighted limitations with ANN is, training likely to stops at local minima. The standard training algorithm to adjust the weights and bias of ANNs is Backpropagation (Shayeghi, Shayanfar, & Azimi, 2009). Nevertheless, Backpropagation is a gradient search algorithm and these kind of training algorithms tend to get trapped at local minima (Sarangi, Singh, Swain, Chauhan, & Singh, 2009; Shayeghi et al., 2009) and are sensitive to the random initial parameters (Bashir & El-Hawary, 2007).

To overcome the above limitations, meta-heuristic optimization techniques are used to train ANNs: Genetic Algorithm (GA) and Particle Swarm Optimization (PSO). GA and PSO are population-based global optimization techniques which are used to solve complex nonlinear objective functions (Subbaraj & Rajasekaran, 2008). The ability of training ANNs with GA and PSO for electricity forecasting has been highlighted in many research (Jeenanunta & Abeyrathn, 2017; Jeenanunta & Abeyrathna, 2017; Mishra & Patra, 2008).

Although the attempt of applying meta-heuristic techniques solve the problem up to a considerable level, the training error, which decides the forecasting performances, can be further reduced by combining their individual properties. Therefore, this research brings a new training algorithm which has both GA and PSO properties to train the ANN for STLF. GA operations are applied to perturb the particles of PSO when it traps in local minima. This helps PSO to create a completely different particle set while they keep the previous personal best and global best values. In case newly created particle can beat the current personal best or the global best, these values are updated and carried to the next generation.

The rest of the paper is arranged as follows: First, the basics of the ANNs and training algorithms (Backpropagation, GA, PSO, and PSO+GA) are explained. The data cleaning process, data arrangement, and training the suggested ANN with different training algorithms are explained under the Design of Experiment. Finally, the conclusion is made after discussing the results of the experiment.

ARTIFICIAL NEURAL NETWORKS AND TRAINING ALGORITHMS

Many researchers (Demiroren & Ceylan, 2006; Krunic, 2000; Senjyu, Tamaki, Takara, & Uezato, 2002) use ANN for electricity load demand forecasting since its ability of recognizing and learning inherent patterns in historical data. As a causal forecasting technique, many external series can be included in ANN such as temperature and time variables. The basic structures of ANNs consist of input nodes, output nodes, and hidden layers. Input nodes are given to represent the external inputs to the network. Output nodes represent the output from the ANNs or the ANNs' results. Hidden layers consist with neurons and all the layers are connected by their specific weights from one neuron to another. When the number of hidden layers and neurons in the hidden layers are increased, the number of weights in the structure are also increased. The weights that connect each node are adjusted during the training phase. The job of the training algorithm is adjusting the weights of the network so that it gives the minimum error between the network output for the training inputs and the target data. The weights adjusting procedure is different from one training algorithm to another.

Backpropagation is the most common training method among researchers (Demiroren & Ceylan, 2006; Rumelhart, 1986; Senjyu, Tamaki, Uezato, Tomonobu, 2001; Shayeghi et al., 2009). 'Backward Propagation of Errors' has been abbreviated to Backpropagation. Backpropagation adjusts weights based on the given target series. Therefore, this method is called as supervised training algorithm.

Algorithm starts giving random values for weights. Inputs pass through these weights and hidden neurons to get the network outputs. These network outputs are compared with the given target values to calculate the errors. These errors are propagated back to the previous layers and during the process weights are adjusted to minimize the errors. This process runs until the output error becomes less than a predetermined value or stop after running predetermined number of cycles (epochs). After stopping the weights adjusting process, the network is ready to work with unseen data.

Genetic algorithm is used to solve complex optimization problems. This method is based on the natural selection or it has been derived from biological evolution (Survival of the Fittest) which was first introduced by Charles Darwin (Sarangi et al., 2009). GA starts with a random population. This population includes a set of individual solutions (Chromosomes). Random values for variables of the objective function bring different objective values or solutions. These random values in chromosomes are called as Genes. Then, GA calculates fitness values of each chromosome. The best chromosomes based on their fitness values are selected as parents. GA uses these parents in the current population to produce children for the next generation. This is how GA expects to get better and better fitness values over the generations. Over successive generations, chromosomes in the populations evolve toward an optimal solution.

PSO has some similarities with GA. Both methods start with random solutions for the given objective function (fitness function). In PSO, individual solutions in the population are called particles. Both methods check the conditions of each individual by calculating the fitness value of them. Based on the calculated fitness values, they produce the next population until they meet the stopping conditions. But PSO does not have evolution operators like in GA: producing crossover and mutation children. Its individuals follow the best particles from their solution space and run until they reach to the target. Since it has few parameters to adjust, it is easier to use and control than GA (Shayeghi et al., 2009).

Generally, PSO shows poor performances as it still has a chance to trap in local minima in the solution space. Since all the particles follow its own personal best and the global best particle within certain generations, all particles would trap at particular solutions. However, several researchers have attempted to combine PSO with some other techniques to increase its performance: Leboucher et al. combine PSO with Evolutionary Game Theory to solve a resources allocation problem (Leboucher, Chelouah, Siarry, & Le Ménec, 2012), Kumar and Singh combine PSO with Bacterial Foraging Optimization for Biometric Feature Optimization (Kumar & Singh, 2017), and Hein and the team apply reinforcement learning with PSO in their research to solve complex optimization problems

(Hein, Hentschel, Runkler, & Udluft, 2016). Concurrently, we combine PSO with GA to train ANNs and the detailed producer is given in the next section.

Research discussed in the previous sections are summarized in Table 1. For instance, we found different STLF techniques in the literature and out of them one research uses regression and three use Deep Learning and so on.

In addition to the STLF techniques, Table 1 presents the techniques that used to compare the results of ANNs, limitations of ANNs, training algorithms of ANNs, and other techniques used with PSO to create hybridize algorithms.

DESIGN OF EXPERIMENT

Data Cleaning

The research is arranged to forecast the days in 2013 using one-year training data in the training phase. Therefore, a sample data set (1st January 2012 to 31st December 2013) is selected from the data gathered by Electricity Generating Authority of Thailand. However, abnormal behavior of data due to holidays, bridging holidays, and outliers reduce the forecasting accuracy. Therefore, the sample data set is cleaned before use them to train the ANN. At the beginning, calendar holidays and bridging holidays which is a day between two holidays or between a holiday and a weekend are replaced by the same-day-of-the-week from previous two weeks as given in Equation (1). Only the latest two weeks are used to avoid the unnecessary variation in the patterns from old data:

$$L^{p}{}_{t}\left(d\right) = w_1 L_t\left(d-7\right) + w_2 L_t\left(d-14\right) \tag{1}$$

where, $L_t^p\left(d\right)$ is the estimated demand and $L_t\left(d\right)$ is the actual load demand at time *t* on day *d*. Giving the priority for the most recent week, w_1 and w_2 are optimally set to 0.7 and 0.3, respectively.

Data is separated into different time windows considering the similar load behavior of the same time period of the same-day-of-the-week. Four period moving average (k=4) and the standard deviation of each time window is used to create the Time-Window based filtering band for each time window as given in Equation (2). All the data outside the filtering band are identified as outliers:

Table 1. Summary of the previous attempts

Concern	Details (The Number of Attempts)					
STLF techniques	Regression (1)	Deep Learning (3)	Stochastic Time Series (1)	Fuzzy Logic (1)	Machine Learning (1)	ANNs (5)
Compare ANN with	Box-Jenkins	Support Vector Machines				
Limitations of ANN	difficulty of determining parameters	requires large computational times	Performances rely on the random initial values	Training likely to stops at local minima		
ANNs' training algorithms	Backpropagation (4)	Genetic Algorithm (1)	Particle Swarm Optimization (2)			
Hybridize PSO with	Evolutionary Game Theory	Bacterial Foraging Optimization	Reinforcement Learning Approaches			

$$B_t\left(d\right)=\frac{\left[\sum_{i=1}^{k}L_t\left(d-7\times i\right)\right]}{k}\pm N\times SD\left(V_t\left(d\right)\right);\quad t=1,\ldots 48 \tag{2}$$

where:

$$V_t\left(d\right)=\left[L_t\left(d'\right),L_t\left(d'-7\right),L_t\left(d'-14\right),\ldots L_t\left(d'-7\times m\right)\right]$$

where:

$$d\in\left\{d',d'-7,\ldots,d'-7m\right\},d'\,is\ the\ last\ 7\ days\ in\ the\ data\ set \tag{3}$$

and $B_t\left(d\right)$ indicates the Time-Window based filtering band of day *d,* at time period *t*. The *N* defines the size of the band width and $V_t\left(d\right)$ is the Time-Window containing all loads of the same period *t* that also include $L_t\left(d\right)$. $SD\left(V_t\left(d\right)\right)$ is the standard deviation of all periods in Time-Window $V_t\left(d\right)$. *m* of Time-Window $V_t\left(d\right)$ is the number of weeks and equal to 105 when the day of week is Sunday, Monday, and Tuesday. *m* is equal to 104 for all the other vectors.

Outliers identified by the Time-Window based technique are replaced using the Equation (1) according to the following conditions:

$$L^p{}_t\left(d\right)=\begin{cases}w_1L_t\left(d-7\right)+w_2L_t\left(d-14\right), & if\ L_t\left(d\right)\notin B_t\left(d\right)\\ L_t\left(d\right), & otherwise\end{cases} \tag{4}$$

Data Arrangement

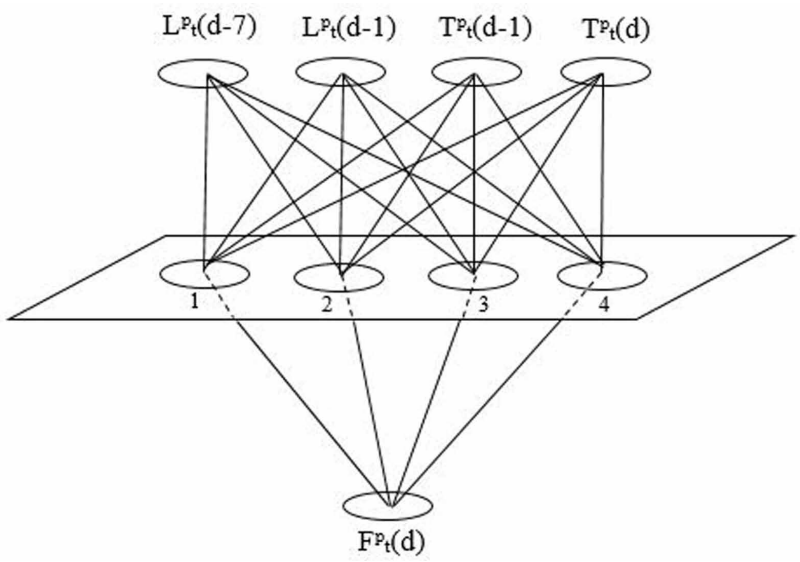
A sample data set is arranged in segment to forecast each time period *t* of each day *d* separately for the year, 2013. The similar load behavior of the same time periods of the previous day and the same-day-of-the-week is considered. Also, the concept of similar temperature behavior of the same time period of the previous day and the forecasted temperature values for the forecasting day are used. One-year data is used to train the network each time for forecasting each time period of each day in 2013. The notion of walking-forward testing routine is applied to forecast each time period where the same amount of training sets is used while forecasting each time period of each day. Considering Monday as an example, for forecasting the time periods of Mondays in 2013, the ANN is trained with one-year data so that only Mondays are in the target series, which gives only 52 training sets to forecast each Monday in 2013. Likewise, 52 training sets are used to forecast each Sunday in 2013 and 51 training sets for forecasting all the other days.

Training the ANN

According to the data arrangement, electricity demand values from two days $\left(L^p{}_t\left(d-7\right)\right)$ and $L^p{}_t\left(d-1\right)$) of the same time period *t* and temperature values from two days ($T^p{}_t\left(d-1\right)$ and $T^p{}_t\left(d\right)$) of the same time period *t* are given as inputs to the network as given in Figure 1.

One hidden layer with four neurons is added to the network. The network has 16 weights to connect input nodes and hidden neurons and 4 weights to connect hidden neurons and the output

Figure 1. The suggested ANN structure



node. All hidden neurons and output nodes have bias. Therefore, 5 bias can be found in the network (4+1). All these weights and bias are variables and adjusted during the training phase. Therefore, total of 25 (16 + 4 + 5) variables can be found from the suggested network. This suggested ANN structure is illustrated by Figure 1 $L^p_t\left(d\right)$ represents the cleaned load of day *d*, at time *t*. Likewise, $T^p_t\left(d\right)$, and F^p_t (*d*) represent cleaned temperature and forecasted load of day *d*, at time *t*, respectively.

Train With Backpropagation

When the network and the data set is ready, backpropagation algorithm start adjusting the weights and bias of the network. Only the stopping conditions have to be set. These stopping conditions are namely the minimum error (MSE) between the network outputs and the target series and the maximum number of epochs that the algorithm should run. Until one of these conditions are met, backpropagation algorithm adjusts the weights and bias based on the current error. Therefore, expecting the maximum performance, the minimum or the target error is set to zero and the maximum number of epochs is set to 1000.

Train With GA, PSO, and PSO+GA

Encoding

Before training the weights and bias of the ANN using GA and PSO, all the weights and bias have to be taken into chromosomes and particles. These weights and bias represent specific genes in the chromosomes and particles. According to the suggested structure in the previous section, 25 genes should have in each chromosome and particle. First 16 genes of the chromosomes and particles belong to the weights that connect input nodes and neurons of the hidden layer. The bias of the hidden neurons ($b_{(1,1)}$ - $b_{(1,4)}$) are placed just after the first 16 weights. Since there is 1 output node, there are 4 weights to connect hidden neurons and the output node. These weights are placed after the bias of the hidden neurons and then the bias of output node ($b_{(2,1)}$) is placed.

Fitness Values Calculation

Mean Squared Error (*MSE*) between the target series $\left(L^{p}{}_{t}\left(d\right)\right)$ and the training outputs $\left(F^{p}{}_{t}\left(d\right)\right)$ is used as the fitness function. Therefore, weights and bias are adjusted and updated at every generation while minimizing the error between the target and training output series. *MSE* between target series and the training outputs is given by:

$$MSE^{d}{}_{t} = \frac{1}{n}\sum_{s=1}^{n}\left(L^{p}{}_{t}\left(d-7s\right)-F^{p}{}_{t}\left(d-7s\right)\right)^{2}, \quad for\ n\ training\ sets \tag{5}$$

n represents the number of data sets used to train the network or simply the number of days we have given in the target series for the specific time period *t*. Since each time period is forecasted separately, *t* varies from 1 to 48 with a new training data set each time to forecast a one complete day. Likewise, fitness vales of all the chromosomes and particles in the current generation are calculated.

Parameters and Functions in GA

After calculating the fitness values of all the chromosomes, GA ranks chromosomes based on their fitness values. Since GA wants to minimize the fitness value, chromosomes with the lowest fitness values are selected as the parents. In this research, there are 100 chromosomes in the population and best 30% is selected as the parents. The composition of the next generation is 5% from elite children, 50% from the crossover children, and 45% from the mutation children.

The top 5 chromosomes of the ranked parents' list are selected as elite children. Without making any change in them, these chromosomes are directly included to the next generation. Two parents from the selected list are selected randomly for producing the crossover children. Genes from the first parent are selected up to a random position and the remaining genes are selected from the other parent for generating the crossover children. This process runs until it creates 50 crossover children. Total of 45 mutation children are produced by randomly selected parents. As each chromosome has 25 genes, 5 randomly selected genes are changed by adding random values from the Gaussian distribution. The process of generating new populations is terminated when GA reaches the maximum generations of 200, the minimum fitness of 0, or the stall generations of 20 where there is no or minor improvement of the fitness value over successive generations.

Parameters and Functions in PSO

The number of elements in each particle is already defined when the ANN structure is made. The number of particles in a population are varied on how difficult the problem is. This research sets 100 particles for a generation. For updating the velocity of each particle for the next generation ($V^{i}\left(t+1\right)$) as given in Equation (6), the inertia weight *w*, random values *rand1* and *rand2*, and the learning factors *c1* and *c2* have to be set. The inertia weight is set to 0.6 so that the current velocity has a higher impact on the velocity of the next generation. Two random values between 0 and 1 created and assigned to *rand1* and *rand2*. Learning factors are set to 2 and 0.5, respectively so that each particle gets higher impact from the personal best than the global best. The updated velocity is used to update the position of each particle ($X^{i}\left(t+1\right)$):

$$V^{i}\left(t+1\right) = w\times V^{i}\left(t\right)+c1\times rand\ 1\times\left(p-X^{i}\left(t\right)\right)+c2\times rand\ 2\times\left(g-X^{i}\left(t\right)\right) \tag{6}$$

$$X^{i}\left(t+1\right) = X^{i}\left(t\right)+V^{i}\left(t+1\right) \tag{7}$$

Same as in GA, PSO algorithm is terminated when it reaches the maximum generations of 200, the minimum fitness of 0, or the stall generations of 20.

Apply GA on PSO

When the fitness value shows minor improvements over successive generations, we assume that it has reached to a local minimum or the global minimum and stars counting the stall generations. When stall generations reach to 20, GA operations are applied to perturb the particles of PSO. Since the size of the population is 100, fifty mutation and fifty crossover children are produced. Fifty particles from the last generation are selected randomly for producing the mutation children. Random values between -1 and 1 are added to the randomly selected 5 elements of each selected particle. For producing crossover children, the same method applied in GA is used where a new particle is created using a random number of genes of a randomly selected particle from the previous generation and remaining genes from another randomly selected particle. The phases of the proposed algorithm are highlighted in the following flowchart (Figure 2).

GA operations help PSO to create completely a different particle set while they keep the previous personal best and global best values. In case newly created particle can beat the current personal best or the global best, these values are updated and carried to the next generation. After applying these GA operations, PSO starts performing its regular activities for finding the global best. Likewise, GA operations are applied two times in PSO over one training period in order to minimize the training error without stopping it at a local minimum.

Parameters used at each training algorithm are summarized in Table 2. Since the same ANN was used at each phase to train with different training algorithms, variables, which should be optimized, are equal to for all training algorithms and their target training error is 0.

Evaluating the Accuracy

Forecasting accuracy of the testing data set is presented in terms of the Mean Absolute Percentage Error (MAPE) for each day, separately (48 values at a time) as in Equation (9). Forecasted outputs are denoted as $F_t^p\left(d\right)$, where *t* is the time intervals of day *d.* These forecasted series are compared with actual demand series after cleaning the data, $L_t^p\left(d\right)$. Therefore, the error series can be calculated as given below:

$$e_t^p\left(d\right) = L_t^p\left(d\right) - F_t^p\left(d\right) \tag{8}$$

Figure 2. Flowchart which connect the phases of the PSO+GA training algorithm

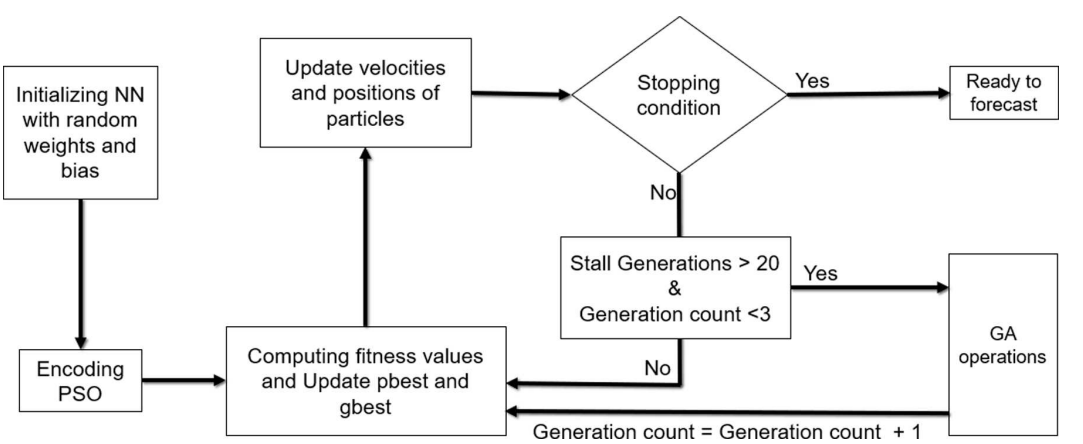


Table 2. Training parameters of each training algorithm

Parameter	Backpropagation	GA	PSO	PSO+GA
No. of ANN variables	25	25	25	25
Target Training Error	0	0	0	0
Training Epochs/ No. of Generations	1000	200	200	200
Population	-	100	100	100
No. of Stall Generations	-	20	20	20
Elite children	-	5%	-	-
Crossover Children	-	50%	-	50%
Mutation Children	-	45%	-	50%

Then the MAPE for day *d* is calculated by the following equation and denoted as $MAPE_d^p$:

$$MAPE_d^p = [\left(\frac{1}{48}\right)\sum_{t=1}^{48}\left|\frac{e_t^p\left(d\right)}{L_t^p\left(d\right)}\right| \times 100] \tag{9}$$

RESULTS AND DISCUSSION

Table 3 gives the monthly average $MAPE^p$ of 15 runs for different training algorithms. Average for each column has been calculated to get the average $MAPE^p$ for the year 2013 by each training algorithm. Average yearly $MAPE^p$ helps to select the best training algorithm to train the ANNs for forecasting short term load demand. Therefore, considering the yearly average $MAPE^p$, the training algorithm which combines GA and PSO methods performs better compared to the other training algorithms. The yearly average $MAPE^p$ with the PSO+GA algorithm is 2.867 and this is the smallest $MAPE^p$ compared to other yearly $MAPE^p$: 3.108 with GA, 3.430 with PSO, and 3.759 with Backpropagation.

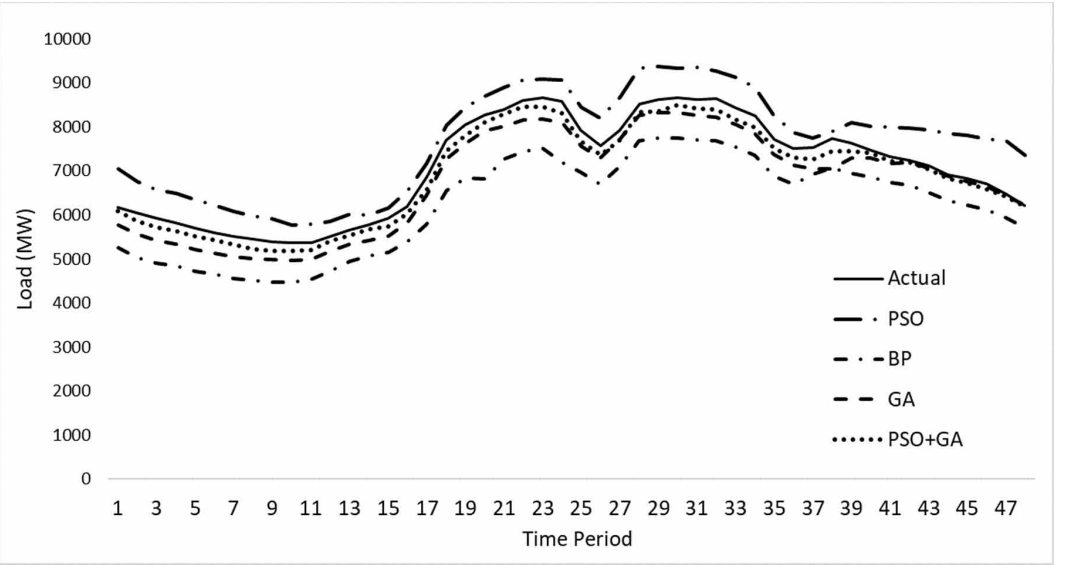
The minimum and the maximum monthly average $MAPE^p$ with the hybridized algorithm are 2.164 (April) and 6.761 (December), respectively. These values when GA is used with the ANN are 2.442 (September) and 7.069 (December). The minimum (2.556) and maximum (8.636) monthly average $MAPE^p$ with PSO training algorithm belong to November and December, respectively. The Backpropagation training algorithm which gives the highest yearly average $MAPE^p$ has its minimum monthly average $MAPE^p$ in August (2.541) and the maximum monthly average $MAPE^p$ in December (8.466).

Only December has higher than 3.00 average monthly $MAPE^p$ when the ANN trains with the hybridized training algorithm. Average monthly $MAPE^p$ s are higher than 3.00 in February, March, June, and December when ANN uses GA. January, February, March, June, October, and December have over 3.00 average monthly $MAPE^p$ s with PSO while only August, September, and November give less than 3.000 $MAPE^p$ values with backpropagation. The difference of the forecasts over the actual demand of a sample day (24th of April, 2013) is given in Figure 3.

Table 3. Monthly (2013) average $MAPE^p$ for different training algorithms

	Monthly Average $MAPE^p$			
	PSO+GA	GA-ANN	PSO-ANN	BP-ANN
January	2.638	2.926	3.137	3.836
February	2.836	3.040	3.215	3.479
March	2.886	3.104	3.494	4.254
April	2.164	2.491	2.603	3.394
May	2.411	2.643	2.938	3.416
June	2.843	3.112	3.202	3.594
July	2.392	2.612	2.931	3.172
August	2.173	2.453	2.573	2.541
September	2.265	2.442	2.611	2.655
October	2.746	2.929	3.259	3.388
November	2.293	2.470	2.556	2.917
December	6.761	7.069	8.636	8.466
Average	2.867	3.108	3.430	3.759

Figure 3. Actual vs. Forecasted load with different training algorithms for April 24, 2013 (Wednesday)



Regardless of the training algorithm, average monthly $MAPE^p$ s in December are higher compared to the other months. The reason is, although holidays, bridging holidays, and outliers from the sample data set are replaced with the estimated data, still the load consumption in December is much lower compared to the other months. When the ANN is trained with higher consumption values from other months, it forecasts assuming that December has similar consumption values.

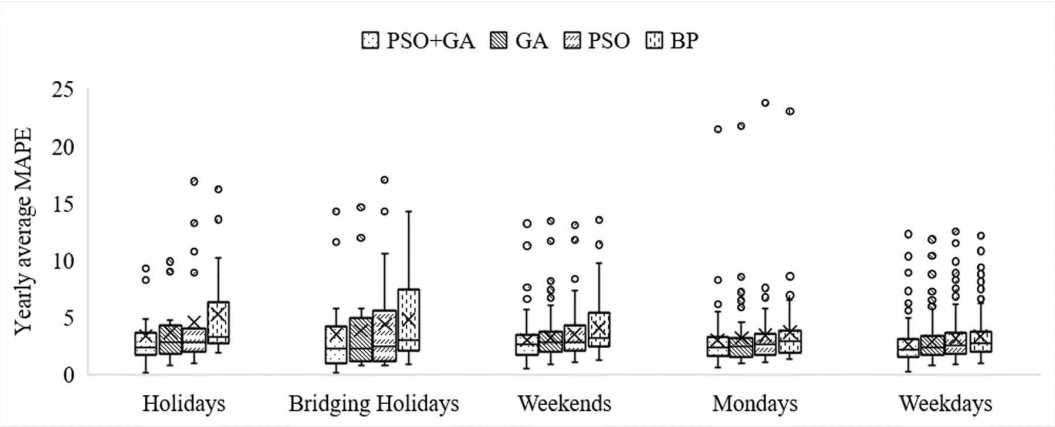
Figure 4 summarizes the yearly average $MAPE^p$ s of different categories of days: Weekdays, Mondays, Weekends, Holidays, and Bridging holidays with different training algorithms. The best yearly average $MAPE^p$ for all the categories comes with the hybridized training algorithm, while the maximum yearly average $MAPE^p$ for all the categories belongs to backpropagation training algorithm. Considering the expansion of boxes and maximum $MAPE^p$ s of each category, promising results can be expected for Weekdays, regardless of the training method. However, the average $MAPE^p$ s and error expansions are higher when the network forecasts Holidays and Bridging Holidays. The highest maximum errors come with the Monday forecasting, regardless of the training method.

CONCLUSION

The objective of the research is to identify the best way of training ANNs for improving forecasting outcomes by minimizing the training error. Due to the limitations of using the backpropagation training algorithm, where it can trap in local minima, two meta-heuristic techniques (GA and PSO) are suggested for further reducing the training error. Another training algorithm is created using the qualities of both GA and PSO techniques. PSO cannot get out of the local minima in case it traps in one. When it traps in a local minimum, crossover and mutation functions of GA are introduced to perturb the positions of each particle in the solution space and to help getting out from the local minimum. These theoretical concepts have been proved with the arranged experiments for Short Term Load Forecasting for the case of Thailand. According to the daily forecasting results for 2013 with the suggested ANN, the best training algorithm to get a good forecasting outcome is the hybridized training algorithm (PSO+GA). GA also gives competitively good results compared to the hybridized training algorithm. However, PSO and Backpropagation training algorithms are fairly good compared to the other two training algorithms.

Regardless of the training algorithm, the hardest month to forecast is December. Due to the low electricity consumption in December, all the forecasting outcomes are higher than the actual electricity

Figure 4. Box plot of yearly average $MAPE^p$ s of different categories of days by different training algorithms



consumption. Other than December, February, March, and June are also difficult to forecast according to the monthly average $MAPE^p$ of those months.

Days of the sample set are categorized into five sections based on the patterns of the consumption curves. Out of them, forecasting results of weekdays except Mondays are promising. Therefore, the research can be further extended to improve the forecasting outcomes of the other categories: Mondays, Weekends, Holidays, and Bridging holidays. Also, some strategies have to be used for forecasting the demands in December. These could be achieved by creating new training algorithms, new ANN feature arrangements, and/or with new data arrangements which help to capture the seasonal variations of data.

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